

An Integer-Floating-Point Dual-Mode Gain-Cell Computing-in-Memory Macro for Advanced AI Edge Chips

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Abstract—This article presents a novel integer-floating-point (INT-FP) gain-cell (GC)-computing-in-memory (CIM) structure for high-precision multiply-and-accumulate (MAC) operations with high computational flexibility, energy efficiency, and inference accuracy. The proposed device employs: 1) a dual-mode zone-based input processing scheme (ZB-IPS) aimed at eliminating exponent subtraction in order to enhance energy and area efficiency (AEF); 2) a dual-mode local computing cell (DM-LCC) to reuse exponent addition as an adder tree stage for INT-MAC to enhance AEF in both INT and floating-point (FP) modes; and 3) a stationary-based two-port GC array (SB-TP-GCA) to enable concurrent data updates and computation while reducing system-to-CIM and internal data accesses to improve energy efficiency. A 16-nm FinFET 108-kb GC-CIM macro fabricated using 4T gain cells (GCs) achieved energy efficiency of 99.5 TOPS/W in INT-MAC operations involving 128 accumulations of 8b-input, 8b-weight, and 23b-output; and 46.4 TFLOPS/W in FP-MAC operations involving 64 accumulations of BF16-input, BF16-weight, and FP32-output.

Index Terms—Artificial intelligence (AI), computing-in-memory (CIM), gain cell (GC), inference, multiply-and-accumulate (MAC).

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I. INTRODUCTION

COMPUTING-IN-MEMORY (CIM) is a promising approach to overcoming the von Neumann bottleneck [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19] by reducing the movement of data between system memory and computing devices. SRAM-based CIM (SRAM-CIM) enables multiply-and-accumulate (MAC) operations [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40] with short compute latency, high energy efficiency, and fast system-level response times in the performance of inference tasks. However, the FinFET design of SRAM-CIMs imposes a layout blocking region with design rule constraints between compact-rule SRAM and logic-rule circuits increasing the area overhead of the memory cell array. Gain-cell-based CIM (GC-CIM) improves the area and energy efficiency of SRAM-CIMs by providing denser logic-compatible embedded bit cells with lower memory access energy.

Advanced AI edge chips [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51] require high energy efficiency and computational flexibility with inference accuracy sufficient for a variety of applications. Floating-point (FP) numerical representation can be used in complex neural networks (NNs) that require high inference accuracy; however, the energy consumption and memory requirements for parameter storage exceed those of fixed-point integer (INT) numerical representation. Many CIM designs [1], [2], [3], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [40], [41], [42], [43] enable high energy efficiency INT-MAC operations; however, few support FP-MAC operations. INT-CIM can emulate INT/FP dual-mode MAC operations; however, the additional adders and alignment circuits [38], [41], [42], [44], [49], [50] required for exponent computation increase the area overhead, power consumption, latency, and reduced area efficiency (AEF), resulting from the fact that FP-MAC functions are idle during INT-MAC operations.

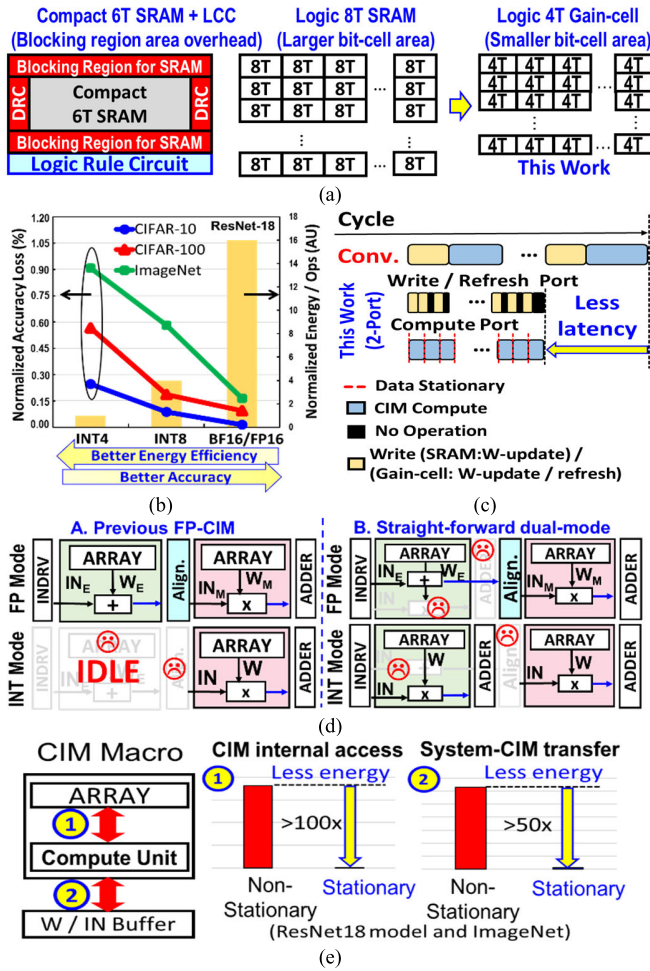


Fig. 1. Challenges in developing GC-CIM. (a) Differences in array area overhead between SRAM bit cells and gain cells under FinFET design. (b) Comparison of integer precision versus floating-point precision in terms of normalized accuracy loss and energy consumption per operation when applied to various datasets. (c) Differences in system-level latency with and without concurrent write-and-compute functionality. (d) Challenges in supporting integer-floating-point dual-mode MAC in CIM macros. (e) High system-level energy consumption resulting from system-to-CIM data transfers.

The fact that existing SRAM-CIMs [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [35], [36], [37], [38], [39], [41], [42] lack support for concurrent write-and-compute functionality greatly increases system-level latency and the occurrence of computation interruptions when updating large quantities of NN data (weight/input), particularly when dealing with low-capacity SRAM-CIMs. Moreover, the system-level energy consumption of NNs is exacerbated by frequent CIM data accesses and the repetitive transfer of data between the system and CIM during computation (see Fig. 1).

The remainder of this article is organized as follows. Section II outlines the structure of the proposed dual-mode gain-cell (GC)-CIM macro and dual-mode local computing cell (DM-LCC). Section III outlines the proposed dual-mode zone-based input processing scheme (ZB-IPS) and toggling-reduced place-value-wise adder tree (TR-PV-ADD) scheme. Section IV outlines the proposed stationary-based two-port GC

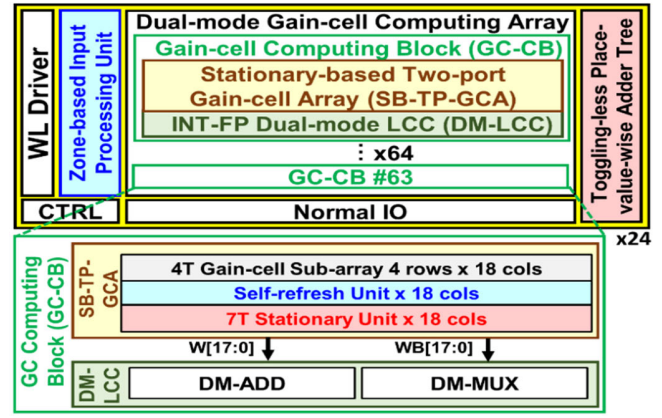


Fig. 2. Structure of proposed dual-mode GC-CIM macro.

array (SB-TP-GCA) scheme. Section V summarizes the performance measurements of the proposed schemes. Section VI concludes this work.

II. DUAL-MODE INT-FP GC-CIM STRUCTURE AND DM-LCC

A. Structure of Proposed GC Dual-Mode CIM Macro

This work sought to overcome the above-mentioned challenges by proposing the following schemes: 1) a dual-mode zone-based input (IN) processing scheme (ZB-IPS) to eliminate the need for subtraction in the exponent (EXP) computation while allowing the use of the alignment circuit in INT-mode to enhance energy efficiency (EEF) and AEF; 2) a DM-LCC, which reuses EXP addition as an adder tree stage for INT-MAC to improve AEF in INT mode; 3) a stationary-based two-port GC array (SB-TP-GCA) with support for concurrent data updates and computation while reducing the frequency of system-to-CIM data transfer and internal data accesses to improve EEF and latency (T_{MAC}); and 4) a TR-PV-ADD based on software-hardware co-design to reduce energy consumption during MAC operations in FP mode. The proposed dual-mode GC-CIM macro performs INT-MAC and FP-MAC operations for general NN applications; i.e., it is not limited to CNNs.

Fig. 2 illustrates the structure of the proposed 108-kb GC-CIM macro, comprising 24 dual-mode CIM banks with output channels. Each bank includes a zone-based IN processing unit (ZB-IPU), a word-line (WL) driver, a dual-mode GC computing array (DM-GCCA), a digital TR-PV-ADD, and a timing controller (CTRL). Each DM-GCCA comprises 64 GC computing blocks (GC-CBs), each of which includes an SB-TP-GCA and an integer-FP (INT-FP) DM-LCC. Each SB-TP-GCA includes a 4T GC sub-array for 72-bit storage data and 18 self-refresh units (SRUs) as well as 18 7T STUs for 18-bit stationary data. Note that the number of GCs can be adjusted during the design phase to ensure a suitable tradeoff between system-to-CIM data transfer energy consumption and macro memory storage capacity versus macro compute AEF and energy efficiency, in accordance with the target application. Each DM-LCC comprises a DM adder (DM-ADD) and eight DM multiplexers (DM-MUXs).

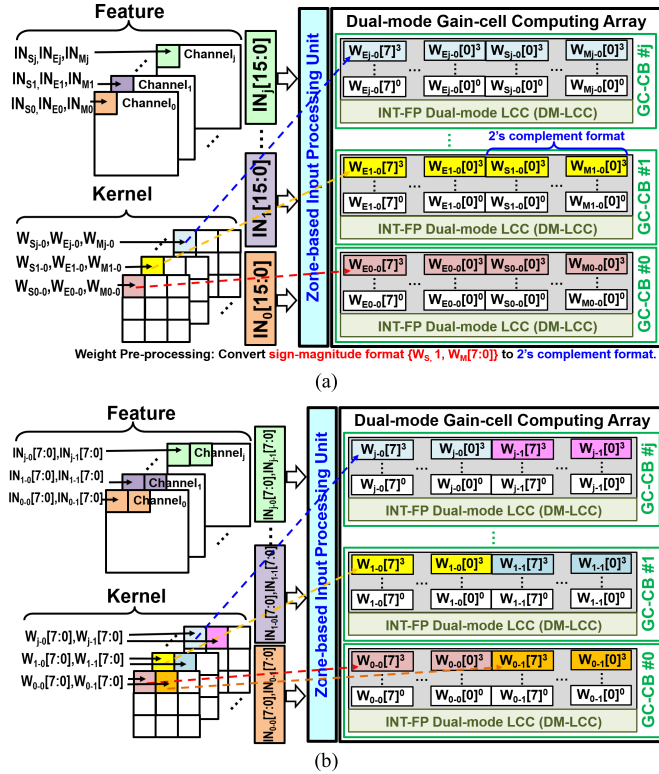


Fig. 3. Weight mapping of (a) j-channel MAC operations with weight data stored in a memory bank in BF16 mode. (b) 2j-channel MAC operations with weight data stored in a memory bank in INT8 mode.

Fig. 3(a) illustrates the proposed bitwise kernel mapping scheme, in which weight data is stored in a dual-mode CIM bank to enable j-channel accumulations in BF16 mode. Each word of BF16-weight data is split into a 1b-sign bit, 8b-weight exponents ($W_E[7:0]$), and 7b-weight mantissa ($W_M[6:0]$). Note that in the BF16 format, a hidden bit prior to the mantissa is implied. Before storing the weight data in the GC array, the sign bit, the hidden bit, and the 7b-weight mantissa are converted to 9b in 2's complement format. Each 8b- W_E and 9b- W_M is stored in 18 4T GCs on the same row across 18 columns of a GC-CB. For example, GC-CB#0 stores a word of 18b- $W_0[17:0]$, while GC-CB#1 stores another word of 18b- $W_1[17:0]$. The weight of each j-channel accumulation is mapped to j GC-CBs in the DM-GCCA.

Fig. 3(b) illustrates bitwise kernel mapping, which is stored in a dual-mode CIM bank for 2j-channel accumulation in INT8 mode. Each pair of words of 8b-weight data is stored in 2's complement format in 18 4T GCs across 18 columns in the same row, wherein GC-CB#0- $W[7:0]$ stores two words of 8b-weight, while GC-CB#1- $W[7:0]$ stores another two words of 8b-weight. Here, the weights for 2j-channel accumulation are mapped to j GC-CBs.

B. Overview of DM-LCC

Fig. 4(a) illustrates the structure of the proposed DM-LCC, which comprises a dual-mode reconfigurable 4b/8b adder in DM-ADD, a leading-1 restoration and sign extension block, and eight dual-mode multipliers. In FP mode,

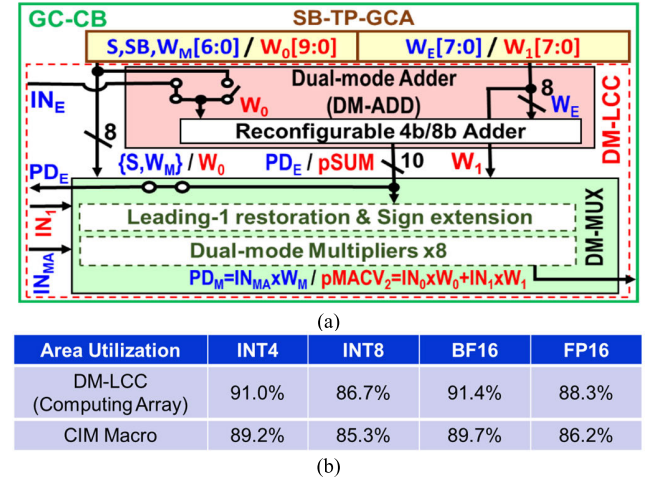


Fig. 4. (a) Structure of proposed DM-LCC. (b) Area utilization of DM-LCC and macro in various modes.

DM-ADD computes the sum of IN_E and W_E , while in INT mode, it computes the pre-compute sum for dual-mode multipliers. DM-ADD can be configured to process two sets of 4b-INT- W_0 + 4b-INT- W_1 for INT4 mode and FP16 mode. Alternatively, DM-ADD can be configured for one set of 8b-INT- W_0 + 8b-INT- W_1 for INT8 mode. DM-MUX processes the leading-1 in FP mode and the sign extension in INT mode. The dual-mode multiplier operates in three modes: 1) FP mode involving the multiplication of 8b aligned- IN_M and W_M ; 2) INT8 mode involving the multiplication of 8b-INT-IN and 8b-INT-W with 2ACCU; and 3) INT4 mode involving the multiplication of 4b-INT-IN and 4b-INT-W with 2ACCU and four output channels. The AEF of the conventional FP-CIM structure in INT mode is low, due to the EXP adders and alignment circuits sitting idle. The dual-mode CIM structure improves energy efficiency and AEF in INT mode by employing the alignment circuit as an IN-sparsity-aware circuit (INAC) and DM-ADDs as an adder tree to enhance N_{ACCU} by $2\times$. Fig. 4(b) outlines the area utilization of the DM-LCC and the macro operating in various modes. The area utilization of the DM-LCC exceeded 86.7% in all modes, while the area utilization of the macro exceeded 85.3% in all modes.

C. Operational Details of Dual-Mode CIM: Integer Mode

Fig. 5(a) illustrates MAC operations involving 128 accumulations of 8b-INT-IN and 8b-INT-W. In INT8 mode, each SB-TP-GCA stores two 8b-INT-W (i.e., $W_0[7:0]$ and $W_1[7:0]$). In phase-1, the DM-ADD sums 8b- W_0 and 8b- W_1 as follows: $pSUM = W_0 + W_1$. As in previous digital-CIM designs [38], [51], [55], this pre-compute sum can be reused for multiple computations by exploiting W data re-use. In phase-2, the ZB-IPU detects IN sparsity to reduce MAC energy consumption in DM-GCCA and TR-PV-ADD. Two bitwise INs (i.e., $IN_0[k]$ and $IN_1[k]$) are then decoded as select signals for the DM-MUX. In phase-3, the DM-MUX performs partial MAC (pMAC) operations for IN_0 and IN_1 to generate the pMAC value ($pMACV = IN_0 \times W_0 + IN_1 \times W_1$). In phase-4, the TR-PV-ADD accumulates 64 pMACVs ($N_{ACCU} = 128$) to output a full-precision 23b MACV.

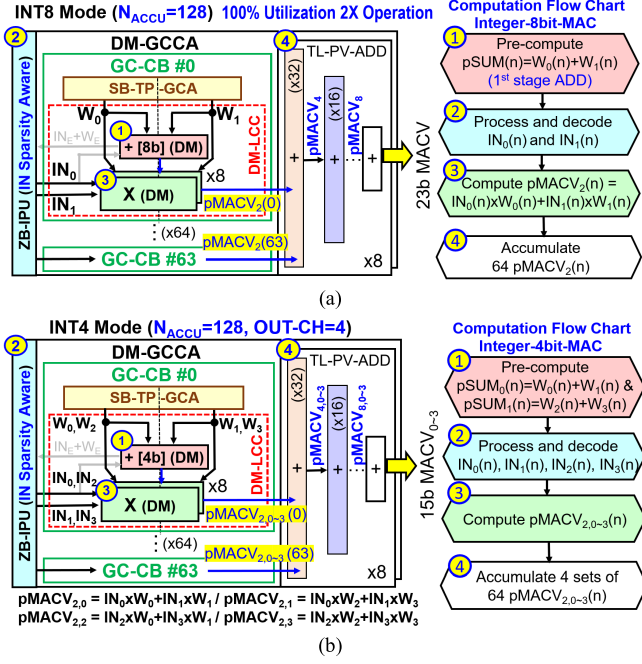


Fig. 5. Computation flow of proposed dual-mode GC-CIM in (a) INT8 mode and (b) INT4 mode.

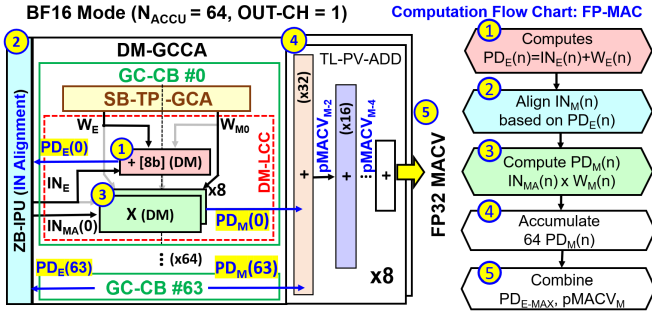


Fig. 6. Computation flow of proposed dual-mode GC-CIM in BF16 mode.

Fig. 5(b) illustrates MAC operations involving 128 accumulations of 4b-INT-IN and 4b-INT-W with four output channels. In INT4 mode, each SB-TP-GCA stores four 4b-INT-W (i.e., $W_0[3:0]$, $W_1[7:0]$, $W_2[7:0]$, and $W_3[7:0]$). In phase-1, the DM-ADD sums 4b- W_0 (W_2) and 4b- W_1 (W_3) to derive two pre-compute sums ($pSUM_0 = W_0 + W_1$ and $pSUM_1 = W_2 + W_3$). In phase-2, the ZB-IPU detects IN sparsity to reduce MAC energy consumption and decodes four bitwise INs (i.e., $IN_0[k]$, $IN_1[k]$, $IN_2[k]$, and $IN_3[k]$) as select signals for the DM-MUX. In phase-3, the DM-MUX performs pMAC operations for IN_0 and IN_1 and generates the four pMACVs with two accumulations ($pMACV_{2,0} = IN_0 \times W_0 + IN_1 \times W_1$). In phase-4, the TR-PV-ADD accumulates four sets of 64 pMACVs ($N_{ACCU} = 128$) to output four full-precision 15b MACV₀₋₃.

D. Operational Details of Dual-Mode CIM: BF16 Mode

Fig. 6 illustrates MAC operations involving 64 accumulations of BF16-IN and BF16-W. In BF16 mode, each SB-TP-GCA stores weight (W) parameters with a 1b-sign (S),

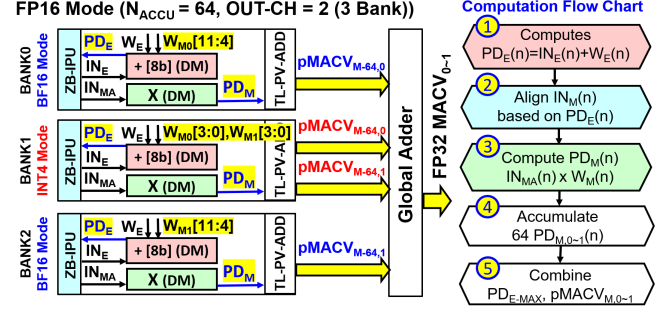


Fig. 7. Computation flow of proposed dual-mode GC-CIM in FP16 mode.

7b- W mantissa (MAN) (W_M) in 2's complement format as well as an 8b- W -EXP (W_E). In phase-1, the DM-ADD sums the 8b-IN-EXP (IN_E) and 8b- W_E to derive the product-EXP (PD_E). In phase-2, the ZB-IPU finds the maximum PD_E (PD_{E-MAX}) and aligns each IN-MAN (IN_M) with the corresponding PD_E to generate an aligned IN_M (IN_{MA}). In phase-3, the DM-MUX computes $IN_{MA} \times W_M$ and generates product-MAN (PD_M). In phase-4, TR-PV-ADD combines 64 PD_M and PD_{E-MAX} with place values to output a full-precision FP32 MACV.

E. Operational Details of Dual-Mode CIM: FP16 Mode

Fig. 7 illustrates MAC operations involving 64 accumulations of FP16-IN and FP16-W. In FP16 mode, three dual-mode CIM banks operate MAC operations through two output channels. Two banks (BANK0 and BANK2) operate in BF16 mode, and one bank (BANK1) operates in INT4 mode. In BF16 mode, SB-TP-GCA stores weight (W) parameters in 2's complement format with a 1b-sign (S), 7b-MSB- W_M ($W_{M0}[11:4]$ and $W_{M1}[11:4]$), and an 8b- W -EXP (W_E). In INT4 mode, SB-TP-GCA stores weight (W) parameters in 2's complement format with two 4b-LSB- W_M ($W_{M0}[3:0]$ and $W_{M1}[3:0]$). In phase-1, the DM-ADD in BANK0 sums the 8b- IN_E and 8b- W_E to derive PD_E . In phase-2, ZB-IPU finds the maximum PD_E (PD_{E-MAX}) and generates IN_{MA} for each IN_M . In phase-3, the DM-MUX computes $IN_{MA} \times W_M$ to generate PD_M . In phase-4, the TR-PV-ADD combines 64 PD_M and PD_{E-MAX} with place values to output full-precision FP32 MACV. In phase-5, the global adder combines 64 PD_M and PD_{E-MAX} with place values to output two full-precision FP32 MACVs (MACV₀₋₁).

III. DUAL-MODE ZB-IPS AND TR-PV-ADD

A. Overview of Dual-Mode ZB-IPS

Fig. 8 illustrates the concept underlying the proposed dual-mode ZB-IPS. In a typical FP-MAC flow [38], [41], [42], [44], [50], the full PD_{E-MAX} bit width is used to compute the number of shifting bits ($N_{SH} = PD_{E-MAX} - PD_E$) as well as the extended MAN alignment bits (exMANb) required to suppress truncation data loss. The use of exMANb in INT mode increases area overhead, resulting in low area utilization.

Fig. 9 illustrates the high-level concept underlying the proposed dual-mode ZB-IPS. Unlike the typical flow in which

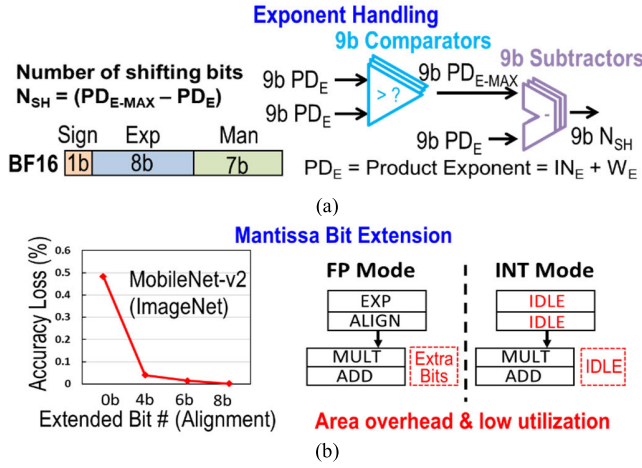


Fig. 8. (a) Large area overhead and power consumption required for exponent handling due to full exponent bit-width processing. (b) High area overhead and low utilization in integer mode, due to mantissa bit extension aimed at suppressing truncation data loss.

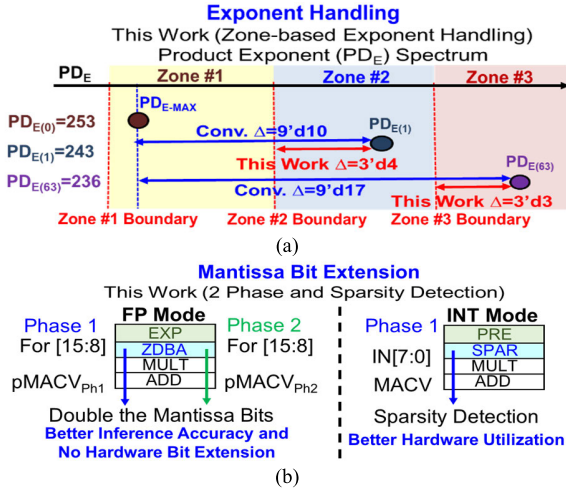


Fig. 9. (a) Comparison of proposed ZB-IPS versus conventional approach in terms of the number of shifting bits. (b) Small area overhead and high area utilization of ZB-IPS with two-phase mantissa alignment.

the mantissa alignment shifting bit ($N_{SH} = PD_{E-MAX} - PD_E$) is derived from the difference between PD_{E-MAX} and PD_E , the ZB-IPS scheme divides the product-exponent spectrum into zones and computes the number of shifting bits based on the difference between product-exponent and its zone boundary. The ZB-IPS adopts two-phase alignment with large exMANb to increase inference accuracy in FP mode while minimizing area overhead to enhance area utilization in INT mode. When applying the ResNet-18 model to the ImageNet dataset, the two-phase alignment of ZB-IPS ensures no loss of accuracy, compared to the software (FP32) baseline while guaranteeing an average arithmetic error of less than 0.1%.

B. Structure of Dual-Mode ZB-IPS

Fig. 10 illustrates the structure of the proposed zone-based input processing unit (ZB-IPU), which comprises 64 DM IN processing blocks (DM-IPBs), a zone bias unit (ZBU), a partial- PD_{E-MAX} finder (pEMAXF), and a zone detector (ZD)

Zone-detect-based alignment (ZDBA)

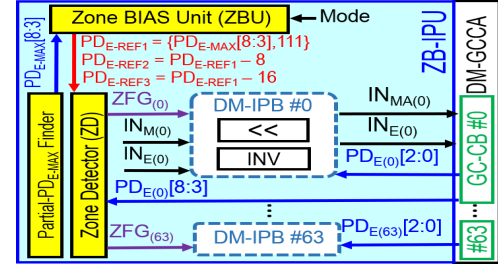


Fig. 10. Structure of ZB-IPU.

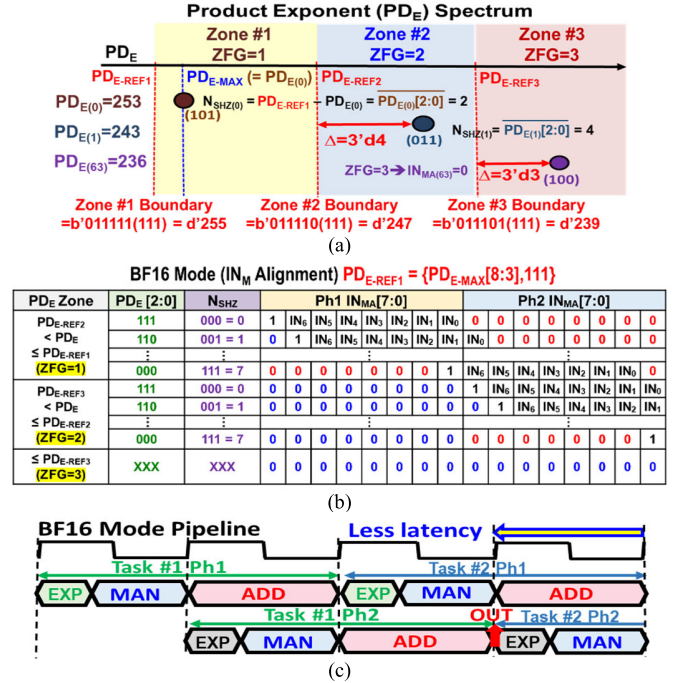


Fig. 11. (a) Operation of ZB-IPU. (b) Operation of two-phase input-mantissa alignment. (c) Timing diagram for two-phase input-mantissa alignment.

for the zone-detect-based alignment (ZDBA) scheme. The pEMAXF derives PD_{E-MAX} based on the 6b MSB instead of the entire exponent, and the ZBU generates three zone references based on the maximum product-exponent. The ZD receives zone references and classifies each product-exponent into one of the three zones, while the DM-IPB computes the zone-shift number and aligns the input mantissa accordingly.

C. Operation of Dual-Mode ZB-IPU

Fig. 11(a) illustrates the operation of the proposed ZB-IPU. The ZDBA scheme includes three stages. In stage-1, pEMAXF finds the MSB-6b ($PD_{E-MAX}[8:3]$) of PD_{E-MAX} . The ZBU then generates three zone references ($PD_{E-REF1\sim3}$) according to $PD_{E-MAX}[8:3]$, namely, $PD_{E-REF1}[8:0] = PD_{E-MAX}[8:3] + 111$, $PD_{E-REF2} = PD_{E-REF1} - 8$, and $PD_{E-REF3} = PD_{E-REF1} - 16$. In stage-2, each $PD_{E(N)}$ is assigned to one of the three zones based on its zone flag (ZFG). In stage-3, the DM-IPB aligns IN_M in accordance with the zone-shift number (N_{SHZ}) obtained by inverting the $PD_E[2:0]$ (LSB-3b), which indicates the difference between PD_E and PD_{E-REF} . ZDBA aligns the

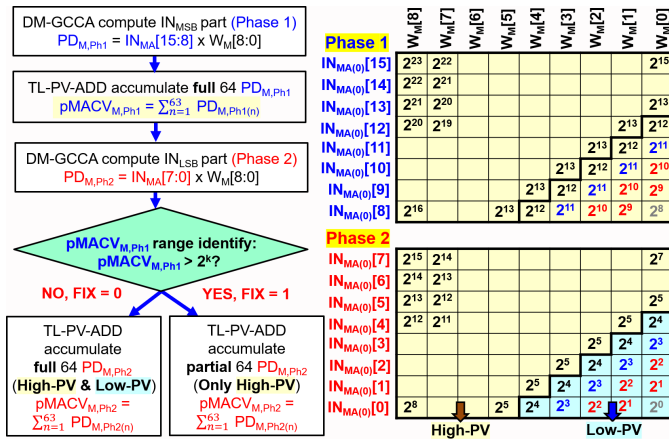


Fig. 12. Operation of TR-PV-ADD.

PD_E of different zones (defined by ZFG) with their respective product–exponent references (PD_{E-REFx}) using the zone-shift number (N_{SHZ}) obtained by inverting PD_E[2:0] (LSB–3b). The elimination of extra physical bit width in alignment circuits and the use of only three small inverters instead of a 9b subtractor to find the shifting bit (N_{SH} = PD_{E-MAX} – PD_E) was shown to reduce the energy and area overhead of the proposed ZDBA to less than those of conventional FP-MAC alignment circuits.

Fig. 11(b) illustrates the alignment of the input mantissa based on its zone and zone-shift number. IN_M alignments are executed for PD_E with $\text{ZFG} = 1$ and 2 in Ph1 and Ph2, respectively. IN_M alignment for PD_E with $\text{ZFG} = 3$ triggers the INAC to reduce compute energy. For example, if $\text{PD}_\text{E-MAX}$ is $\text{PD}_\text{E(0)} = 011111101$ (253) and $\text{PD}_\text{E-REF1} = 011111111$ (255), and then, $\text{PD}_\text{E(0)}$ falls within zone-0 ($\text{ZFG} = 1$) and proceeds to IN_M alignment in Ph1 with $N_{\text{SH(0)}} = 2$ (inversion of $\text{PD}_\text{E(0)}[2:0]$). In the case where $\text{PD}_\text{E(63)} = 236$ ($\text{ZFG} = 3$), IN_M alignment is skipped without data loss (INAC activated) and MAN is zero. Fig. 11(c) presents a timing diagram of two-phase input-mantissa alignment. Pipelining the adder accumulation of Ph1 and the MAN multiplication of Ph2 shortens T_{MAC} . When performing MAC operations in INT8 mode, $\text{PD}_\text{E-REF1}$ is set to 0 and ZD serves as INAC to improve energy efficiency.

D. Toggling-Reduced Place-Value-Wise Adder Tree

Considerable power is consumed by the digital shift and adder (DSaA) used in most digital-domain CIM macros [40], [42], [43], [45], [47]. In this work, we sought to reduce the energy consumption of the DSaA by reducing the rate of signal toggling. Fig. 12 illustrates the operation of the proposed TR-PV-ADD, which is based on a software–hardware co-design aimed at reducing energy consumption during FP-MAC operations. The FP-MAC comprises four steps: 1) multiplying the MSB-8b of IN_{MA} and $8b-W_M$ in DM-GCCA during phase-1; 2) accumulating the product-mantissa (PD_M) in the TR-PV-ADD during phase-1; 3) multiplying the LSB-8b of IN_{MA} and $8b-W_M$ in DM-GCCA during phase-2; 4) identifying the pMACV range in phase-1 ($pMACV_{M,Ph1}$) for comparison with a predefined value of 2^k to determine whether to

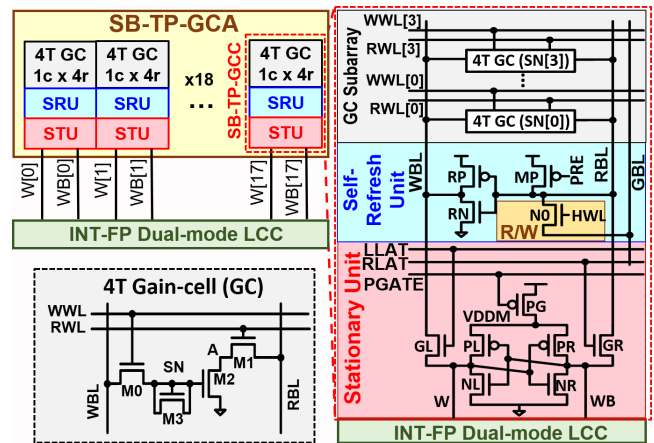


Fig. 13. Structure of SB-TP-GCA and 4T gain cell.

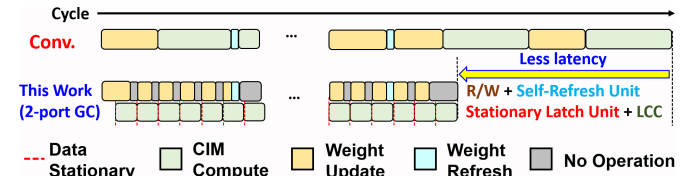


Fig. 14. Operation timing diagram of SB-TP-GCA.

accumulate the full PD_M or the partial PD_M during phase-2; and 5) accumulation of the PD_M in the TR-PV-ADD during phase-2. In step 4, if the $\text{pMACV}_{M,\text{Ph1}}$ exceeds the 2^k value, and then, TR-PV-ADD triggers the termination of computation for the lower PV part during phase-2, which reduces the energy consumption of the TR-PV-ADD by lowering the signal toggling rate. Conversely, if $\text{pMACV}_{M,\text{Ph1}}$ is lower than the 2^k value, and then, TR-PV-ADD proceeds with the full accumulation computation. Adjusting the value of k during the design phase makes it possible to optimize the tradeoff between inference accuracy and power consumption in accordance with the target application.

IV. STATIONARY-BASED TWO-PORT GC ARRAY

A. Overview of SB-TP-GCA

Fig. 13 illustrates the SB-TP-GCA, which comprises 18 stationary-based two-port GC columns (SB-TP-GCCs) operable in three modes: stationary update, storage update, and self-refresh. Each SB-TP-GCC is equipped with four 4T GCs, a 4T SRU, one read/write pass transistor (N0), and one 7T STU.

Fig. 14 presents an operation timing diagram of the SB-TP-GCA. Previous CIMs [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [35], [36], [37], [38], [39], [41], [42] do not support concurrent write-and-compute functionality; however, the proposed GC-CIM supports simultaneous MAC computation and W updating or refreshing to shorten system-level T_{MAC} . Moreover, each SB-TP-GCC uses only 27 transistors for two memory cells (6.75T/cell), which is far fewer than previous two-port (8–12T) CIMs [34], [40], [43], [45].

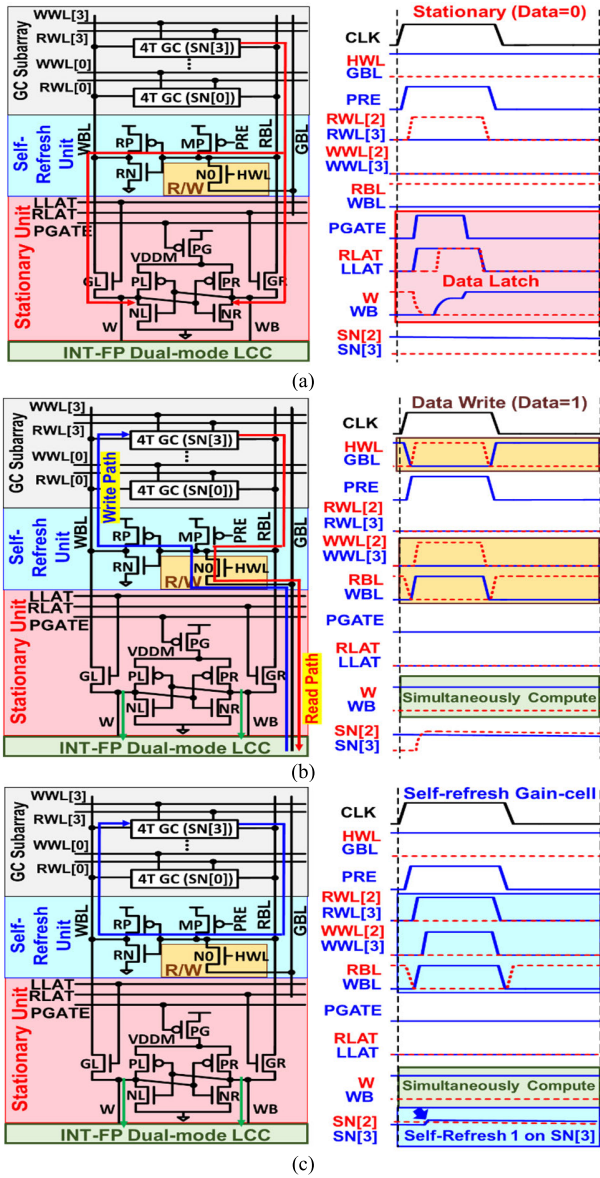


Fig. 15. Operating modes of proposed GC-CIM. (a) Stationary update. (b) Storage update compute. (c) Self-refresh compute.

B. Operational Details of SB-TP-GCA: Stationary-Update Mode

Fig. 15(a) illustrates the stationary-update operation of the proposed GC-CIM. In stationary-update mode, the data stored in the accessed GC ($RWL = 1$) is transmitted to the read BL (RBL) and SRU. By activating the write-assist mode of the STU ($PGATE = 1$, $LLAT = 1$, and $RLAT = 1$), the full-swing differential signal on the RBL and write BL (WBL) drive the accessed data into the STU. This allows the reuse of stationary weight data by DM-LCC over multiple MAC computations. After the STU data are updated, the STU is decoupled from WBL and RBL by setting $PGATE = LLAT = RLAT = 0$, while RBL is precharged to VDD via the MP transistor ($PRE = 0$). The decoupling of the GC sub-array from STU allows the SB-TP-GCA to support simultaneous MAC computation and W updating or refreshing. The low toggling rate generated by the SB-TP-GCA and stationary

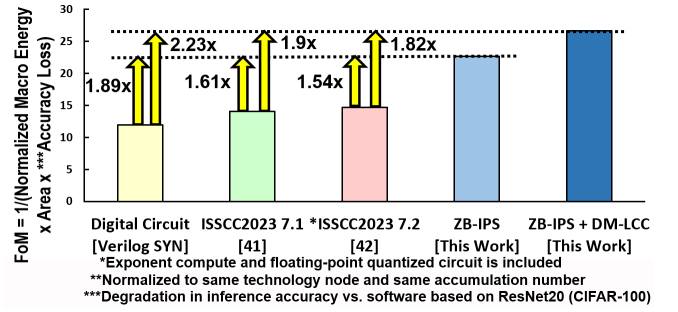


Fig. 16. FoM scores under various exponent computation schemes.

computation flow decreases the energy consumed by the DM-LCC by reducing dynamic transitions, further enhancing macro energy efficiency.

C. Operational Details of SB-TP-GCA: Storage-Update-Compute Mode

Fig. 15(b) illustrates the storage-update-compute and storage-access-compute operations of the proposed GC-CIM. In storage-update-compute mode, write data is passed from the global BL (GBL) to the SRU via the N0 transistor ($HWL = 1$). The inverter (RP and RN) of the SRU then drives the WBL to write the data into the selected GC when the write WL (WWL) is activated. In storage-access-compute mode, the RBL is precharged to VDD via the MP transistor ($PRE = 0$) prior to activating the RWL. When the selected RWL is turned on, the voltage on RBL associated with the storage of data-1 by a GC ($W = 1$) is driven toward the ground. N0 passes the voltage of the RBL to the GBL, whereupon a sense amplifier (SA) detects the GBL voltage to provide a digital readout.

D. Operational Details of SB-TP-GCA: Self-Refresh-Compute Mode

Fig. 15(c) illustrates the self-refresh operation of the proposed GC-CIM. In self-refresh mode, the RBLs are precharged to VDD via the MP transistor ($PRE = 0$) before activating the RWL. When the selected RWL is turned on, the voltage on RBL associated with the storage of data-1 by a GC ($W = 1$) is driven toward the ground. The SRU reads data from RBL and then drives the WBL (as a storage-update operation) to refresh the selected GC with $WWL = 1$. The high bandwidth self-refresh operation of SRUs, which does not require the toggling of the GBL (with its high parasitic resistance and capacitance), reduces the latency and energy consumption associated with refreshing the GC.

V. PERFORMANCE AND EXPERIMENT RESULTS

A. Performance of Proposed Schemes

Fig. 16 compares the proposed ZB-IPS with previous CIM schemes that support exponent computation for FP-MAC. Assessments were based on the figure of merit (FoM) derived from simulation results, as follows: $1/(\text{normalized energy consumption} \times \text{macro area} \times \text{inference accuracy loss})$.

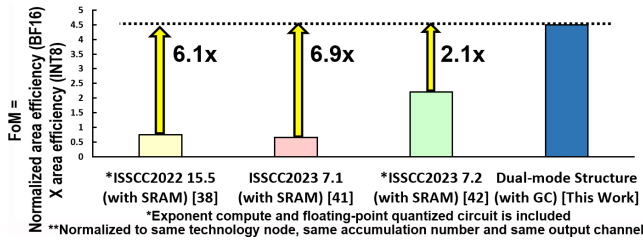


Fig. 17. FoM scores under various CIM structures.

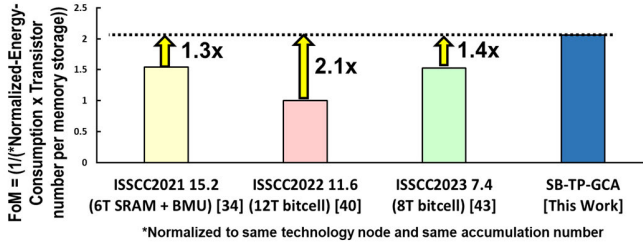
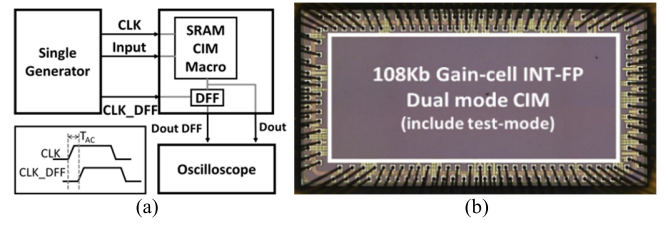


Fig. 18. FoM scores under various CIM works that support concurrent data update and computation.

This analysis deals with the probability distribution of readout values when applying the ResNet20 model to the CIFAR-100 dataset. Previous time-domain schemes [41], [50] used local computing cells to process exponent computation to increase energy efficiency; however, that approach is prone to a loss of computational accuracy due to process variation. Existing digital-domain schemes [38], [42], [44] employ the full PDE_{MAX} bit width in computing the number of shifting bits, which increases the energy consumption associated with exponent computation. The proposed ZB-IPS achieved an FoM score that was 1.5–2.2 $\times$ higher than those of pure-digital and previous SRAM-CIM works, despite the fact that it supports FP-INT dual-mode functionality. The high FoM can be attributed to high inference accuracy resulting from digital-domain computation as well as the low energy consumption resulting from high input sparsity.

Fig. 17 compares the structure of the proposed dual-mode INT-FP GC-CIM with that of previous CIM works. Assessments were based on an FoM derived from simulation results, as follows: normalized AEF in BF16 mode $\times$ normalized AEF in INT8 mode. All works have been normalized to the same technology node, input channel (accumulation number), and output channel to ensure a fair comparison. Note that the exponent compute and FP quantized circuit were included in this assessment. The AEF of previous CIM structures [38], [41], [42], [44], [49], [50] was low, due to the fact that EXP adders and alignment circuits sat idle in INT mode. The FoM score of the proposed DM-LCC and ZB-IPS was 2.1–6.9 $\times$ higher than those of previous CIM works [34], [40], [43] while supporting dual-mode functionality. The high FoM can be attributed to high area utilization and AEF in both INT mode and FP mode.

Fig. 18 compares the proposed SB-TP-GCA with previous CIM works that support concurrent data updates. Assessments were based on an FoM derived from simulation results, as follows: $1/(\text{normalized energy consumption} \times \text{transistor number}$



CHIP SUMMARY				
Technology	16nm FinFET CMOS			
Gaincell Capacity	108Kb			
Input precision (bit)	INT4	INT8	BF16	FP16
Weight precision (bit)	INT4	INT8	BF16	FP16
Number of input channels (N_{ACCU})	128	128	64	64
Number of output channels	96	24	24	16
Output precision (bit)	15	23	FP32	FP32
Supply voltage (V)	0.4–0.8	0.4–0.8	0.4–0.8	0.4–0.8
Access time (ns)	1.9 (0.8V)	1.9 (0.8V)	4.0 (0.8V)	4.0 (0.8V)
Energy efficiency (TOPS/W) (TFLOPS/W) (Average performance)	296.3–397.5	74.3–99.5	34.1–46.4	22.5–29.9
Energy efficiency (TOPS/W) (TFLOPS/W) (*Peak performance)	505.4–658.2	126.6–163.8	72.9–91.8	48.3–59.7
Area efficiency (TOPS/mm ²) (TFLOPS/mm ²)	44.28 (0.8V)	11.07 (0.8V)	2.63 (0.8V)	1.75 (0.8V)

*Performance in 90% input sparsity, 10% input toggle rate (ResNet-20)

(c)

Fig. 19. (a) Timing measurement environment. (b) Die photograph. (c) Chip performance summary.

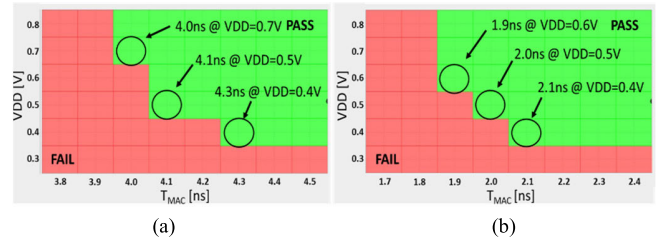


Fig. 20. Measured shmoo plot showing VDD versus compute latency in forward propagation mode under various input-weight configurations. (a) BF16-input, BF16-weight, and FP32-output. (b) 8b-input, 8b-weight, and 23b-output.

per memory storage). The area overhead and energy consumption of the proposed SB-TP-GCA were lower than those of logic-rule SRAM, thanks to a lower transistor count and the absence of current leakage in the memory array. Despite supporting both dual-mode function and stationary computation, the proposed SB-TP-GCA earned an FoM score that was 1.33–2.06 $\times$ higher than those of previous SRAM-CIM works [34], [40], [43] that support only two-port functionality.

B. Experiment Results

The efficacy of the proposed schemes was evaluated using a test chip, which was fabricated with a 108-kb GC-CIM macro using a 16-nm FinFET CMOS logic process. As shown in Fig. 19(a), the test chip was implemented using the DFF-based path-delay exclusion scheme [32], [35], [50] to enable the detection of on-chip MAC-compute latency in the proposed GC-CIM macro. Fig. 19(b) and (c), respectively, presents a die photograph and chip summary table for CNN operations with INT-4b, INT-8b, and BF16 input-weight configurations. Fig. 20 presents measured shmoo plots obtained from the 108-kb GC-CIM test chip. When performing MAC operations involving 64 accumulations of BF16-input and

TABLE I
COMPARISON TABLE OF THE PROPOSED MACRO WITH RECENT SRAM- AND eDRAM-BASED CIM MACROS

Previous ISSCC work	ISSCC 2022 [47]	ISSCC 2022 [37]	ISSCC 2022 [40]	ISSCC 2023 [43]	ISSCC 2023 [52]	ISSCC 2023 [53]	ISSCC 2023 [54]	ISSCC 2019 [4]	ISSCC 2022 [38]	ISSCC 2023 [41]	ISSCC 2023 [42]	This Work			
INT/FP	INT	INT	INT	INT	INT	INT	INT	FP	INT/FP	FP	INT/FP	INT/FP			
Array size	32Kb	1Mb	64Kb	54Kb	9Kb	128Kb	16Kb	16Kb	12Kb	832Kb	64Kb	96Kb			
Technology	28nm	28nm	5nm	4nm	28nm	12nm	22nm	28nm	28nm	22nm	28nm	16nm			
Cell structure	6T +RLPU	6T +EDC	1R1W 12T	8Tx2 + OAI	6T	6T+3T	6T +MU	8T	6T+XOR	EXP: 6T+TD-CU MAN: 6T+DAHLCC	6T+ HFMC/LAMC	4T Gaincell +DM-LCC			
Exponent Structure	-	-	-	-	-	-	-	NMC-Digital	INt: Processor-Digital Wt: Software	IMC - Time	INt: Processor-Digital Wt: Software	IMC-Digital			
Mantissa / INT Structure	IMC-Digital	IMC-Time	IMC-Digital	IMC-Digital	IMC-Analog	IMC-Analog	IMC-Analog	NMC-Digital	IMC-Digital	IMC - Hybrid	IMC-Digital	IMC-Digital			
Input precision (bit)	8	8	8	8	8	8	8	FP32	BF16	FP32	BF16	8	BF16	8	BF16
Weight precision (bit)	8	8	8	8	8	8	8	FP32	BF16	FP32	BF16	8	BF16	8	BF16
Number of input channel	32	64	64	144	16	64	64	-	32	32	128	128	128	128	128
Output precision (bit)	21	22	22	24	20	8	24	FP32	BF16	FP32	FP32	23	BF16	23	FP32
¹ Output-ratio	1	1	1	1	1	0.4	1	-	0.42	0.47	1	1	0.3	1	1
Access time (ns)	24	6.6	22.2	5.4	6	10	11.1	13376	22.8	58.4	6.4	5.5	6.8	1.9	4
Model	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN	CNN
Supply voltage (V)	0.8	0.65-0.9	0.5	0.32-1.1	0.6-0.9	0.5 /0.85	0.72-0.82	0.6-1.1	0.6-1.0	0.6-0.8	0.6-0.9	0.4-0.8			
Average Energy efficiency (TOPS/W) (TFLOPS/W)	27.38	27.75	-	12.3	14.24	-	16.02	0.55	29.2	3.7	16.22	19.5	14.04	99.5	46.4
Peak Energy efficiency (TOPS/W) (TFLOPS/W)	35.55	37.01	*63	87.4	33.44	86.27	21.38	0.55	46.2	5.9	70.2	44	31.6	163.8	91.8
² Norm. Area efficiency (TOPS/mm ²) (TFLOPS/mm ²)	0.5	8.79	*10.69	8.98	2.88	-	3.35	0.171	0.26	0.03	1.3	6.59	**5.26	11.07	2.63
³ Accuracy loss(%) (CIFAR-10) ResNet20	-	-0.12	-	-	-	-	-	-	-	-	-0.03	-	-	-0.02	-0.01
⁴ Accuracy loss(%) (CIFAR-100) ResNet20	-	-0.22	-	-	-	-0.9	-	-	-	-	-0.07	-	-	-0.19	-0.01
⁵ Accuracy loss(%) (ImageNet) ResNet18	-	-	-	-	-	-	-	-	-	-	-0.43	-	-	-0.58	-0.02

¹Output-ratio = Real output resolution/Ideal output resolution

²Normalized to the same technology node (16nm)

*Estimated value

**Exponent compute and Floating-point quantized circuit is not included

³Using ResNet-20 model and the software baseline was 93.13%

⁴Using ResNet-20 model and the software baseline was 70.22%

⁵Using ResNet-18 model and the software baseline was 68.31%

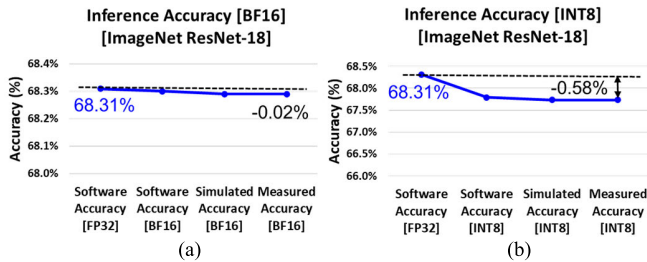


Fig. 21. Measured inference accuracy when applying the ResNet-18 model to the ImageNet dataset under various input-weight configurations. (a) BF16-input, BF16-weight, and FP32-output. (b) 8b-input, 8b-weight, and 23b-output.

BF16-weight with FP32-output, the GC-CIM macro achieved a compute latency of 4 ns at VDD = 0.7 V [see Fig. 20(a)]. When performing MAC operations with 128 accumulations of INT8b-input and INT8b-weight with INT23b-output, the GC-CIM macro achieved a compute latency of 1.9 ns in INT8 mode at VDD = 0.6 V [see Fig. 20(b)]. Table I compares the proposed macro with recent SRAM-CIM/eDRAM-CIM works [4], [37], [38], [40], [41], [42], [43], [47], [52], [53], [54]. Despite using different technology nodes, the proposed dual-mode GC-CIM macro outperformed previous CIM works in terms of energy efficiency and AEF.

Fig. 21 presents experiment results obtained during the application of the ResNet-18 model to the ImageNet dataset. Under the BF16-IN-BF16-W-FP32-OUT configuration, the

inference accuracy was only 0.02% lower than that achieved via software simulation with FP32 precision. Under the 8bIN-8bW-23bOUT configuration, the inference accuracy was only 0.58% lower than the ideal case obtained via software simulation with FP32 precision.

VI. CONCLUSION

This article presents an INT/FP dual-mode GC-CIM macro for CNN operations. The proposed dual-mode ZB-IPS eliminates subtraction from exponent computation to improve energy efficiency and AEF. The proposed DM-LCC reuses exponent addition as an adder tree stage for INT-MAC to improve AEF and area utilization in INT/FP mode. The proposed SB-TP-GCA supports concurrent data updates and computation while reducing system-to-CIM and internal data accesses to improve energy efficiency and latency. The efficacy of the proposed schemes was verified in experiments using a fabricated 16-nm FinFET 108-kb GC-CIM macro. The testchip achieved energy efficiency of 74.3–99.5 TOPS/W for INT-MAC operations involving 8b-input and 8b-weight with 128 accumulation and 23b-output. It also achieved energy efficiency of 34.1–46.6 TFLOPS/W for FP-MAC operations involving BF16-input and BF16-weight with 64 accumulation and FP32-output.

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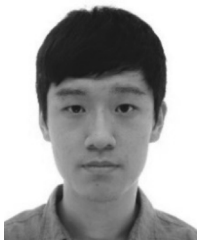
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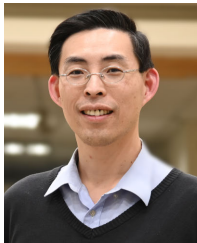


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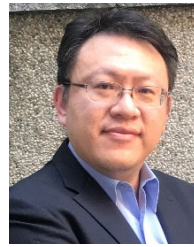
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